TRACTIAN
INDUSTRIAL INTELLIGENCE

HMD10

Industrial I²C Relative Humidity Sensor

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REV A — TFT26 CANDIDATE RELEASE

GENERAL DESCRIPTION

The **HMD10** is a 32-bit resolution digital relative humidity sensor designed for industrial environment monitoring. It communicates over a standard I²C bus and provides humidity data as a 32-bit signed integer in milli-percent RH units, consistent with the data encoding of the companion TMP64 temperature sensor.

The HMD10 shares an identical register layout with the TMP64: register `0x0F` holds a fixed WHO_AM_I identification byte, and a 4-byte sequential read from register `0x00` returns the humidity measurement in MSB-first order.

🔍 **I²C Address: Unknown.**
The HMD10's 7-bit I²C address is factory-configured and not published in this revision. The address must be discovered at runtime by scanning the bus. Valid scan range:
0x08 – 0x77
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FEATURES & BENEFITS

- I²C interface — **7-bit address factory-configurable (unknown)**
- 32-bit signed humidity output in **milli-percent RH**
- Measurement range: 0 %RH to 100 %RH
- Accuracy: ±2 %RH (20–80 %RH), ±3 %RH full range
- WHO_AM_I register at address `0x0F`
- 4-byte sequential read from register `0x00`, MSB first
- Identical register protocol to TMP64
- Open-drain SDA/SCL — multi-device bus compatible
- 1 Hz measurement update rate (nominal)
- Operating voltage: 3.0 V – 3.6 V

I²C ADDRESS

0x??

Factory-configured — must be discovered via bus scan.
Valid range: **0x08 – 0x77** (I²C reserved addresses excluded).

PIN DESCRIPTION

PIN	NAME	TYPE	DESCRIPTION
1	GND	PWR	Ground reference
2	VDD	PWR	Supply voltage, 3.0–3.6 V
3	**SDA**	**I/O OD**	**I²C serial data line — open-drain**
4	**SCL**	**I OD**	**I²C serial clock line — open-drain**
5	ADDR	CFG	Address configuration — set at factory, not accessible at runtime

DC ELECTRICAL CHARACTERISTICS

T_A = 25 °C, V_{DD} = 3.3 V, unless otherwise noted.

PARAMETER	SYMBOL	MIN	TYP	MAX	UNIT
Supply Voltage	V_{DD}	3.0	3.3	3.6	V
Active Supply Current	I_{DD}	—	250	400	µA

PARAMETER	SYMBOL	MIN	TYP	MAX	UNIT
Standby Current	I_{STBY}	—	3	8	µA
SDA/SCL Logic-Low Input	V_{IL}	−0.3	—	0.8	V
SDA/SCL Logic-High Input	V_{IH}	2.1	—	V_{DD}+0.3	V
SDA Output Low Voltage	V_{OL}	—	—	0.4	V
Measurement Range	—	0	—	100	%RH
Humidity Accuracy (20–80 %RH)	**ε_{RH}**	**—**	**±1.5**	**±2.0**	**%RH**
Humidity Accuracy (full range)	ε_{RH}	—	±2.0	±3.0	%RH
I²C Clock Frequency	f_{SCL}	—	—	400	kHz
Measurement Update Rate	f_{CONV}	—	1	—	Hz

I²C TIMING

Figure 1 — Write-then-Read Sequence (Register Read)

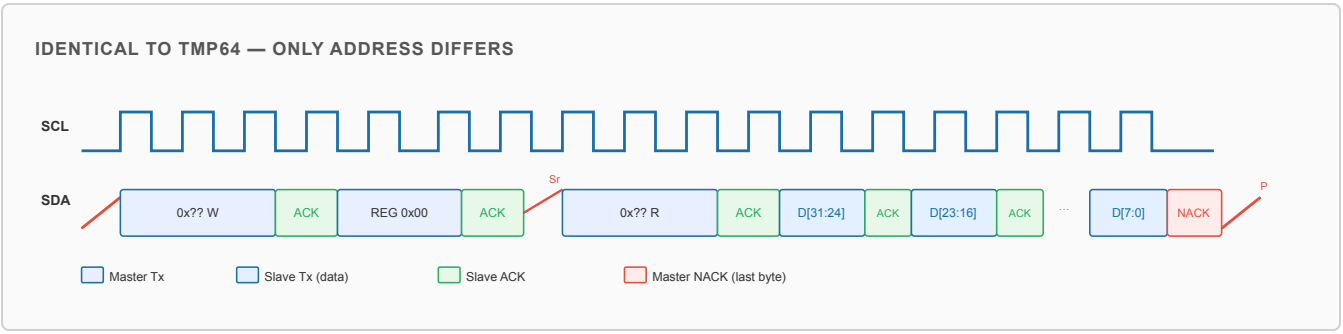
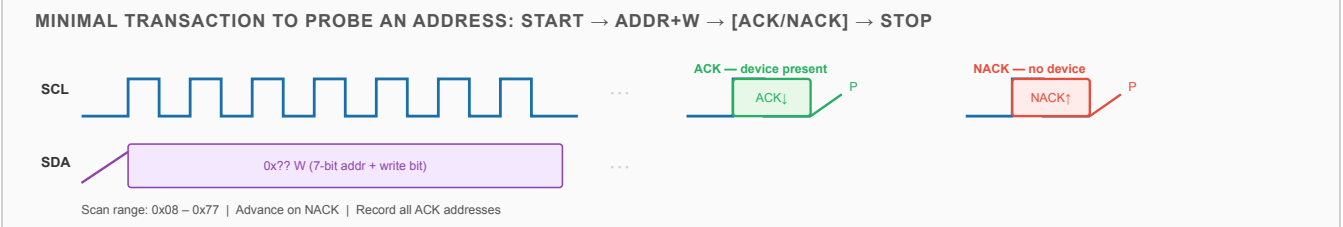


Figure 2 — Bus Scan Probe Sequence



Protocol Symbol Reference

SYMBOL	MEANING	DRIVEN BY
`S`	START — SDA falls while SCL high	Master
`Sr`	Repeated START — no STOP between write and read phases	Master
`P`	STOP — SDA rises while SCL high	Master
`A`	ACK — slave pulls SDA low during 9th clock	Slave
`N`	NACK — master releases SDA high on last byte	Master
`ADDR W/R`	7-bit address + direction bit (0 = write, 1 = read)	Master

SYMBOL	MEANING	DRIVEN BY
`D0…Dn`	Data bytes, MSB first within each byte	Slave

REGISTER MAP

> ℹ **Identical layout to TMP64.**
> Same register addresses and the same 4-byte MSB-first sequential read protocol. Only the I²C address and the physical meaning of the 32-bit value differ.

ADDRESS	NAME	ACCESS	DESCRIPTION
`0x0F`	**WHO_AM_I**	**RO (1 byte)**	**Fixed device identification byte unique to HMD10 family.**
`0x00`	**HUM_MSB**	**RO**	**Humidity bits [31:24]. Sequential read from `0x00` returns all 4 bytes in order.**
`0x01`	**HUM_B2**	**RO**	**Humidity bits [23:16].**
`0x02`	**HUM_B1**	**RO**	**Humidity bits [15:8].**
`0x03`	**HUM_LSB**	**RO**	**Humidity bits [7:0] — least significant byte.**
`0x04–0x0E`	RSVD	—	Reserved. Do not access.
`0x10–0xFF`	RSVD	—	Reserved. Do not access.

0x0F — WHO_AM_I Device Identification

`RO · Value: device-specific`

7–0							
ID[7:0]							

Fixed identification byte unique to the HMD10 device family. Must be read and verified after bus scan to confirm the discovered address belongs to the humidity sensor.

> 🔍 **Value not published in this revision.**
> Read WHO_AM_I from any address that responds during the bus scan and print the returned byte. The simulator will confirm device identity.

0x00–0x03 — HUM[31:0] Humidity Data (32-bit, MSB first)

`RO · 4 bytes sequential`

byte 0	byte 1	byte 2	byte 3				
H[31:24]	**H[23:16]**	**H[15:8]**	**H[7:0]**				

32-bit signed two's complement integer in **milli-percent RH**. Divide by 1000 to obtain %RH. Master ACKs bytes 0–2; master NACKs byte 3 to terminate the transfer.

RAW VALUE (INT32)	HUMIDITY	HEX (32-BIT)
0	0.000 %RH	`0x00000000`
30000	30.000 %RH	`0x00007530`
60000	**60.000 %RH**	**`0x0000EA60`**

RAW VALUE (INT32)	HUMIDITY	HEX (32-BIT)
85500	85.500 %RH	`0x00014E1C`
100000	100.000 %RH	`0x000186A0`

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